Evaluating Hallucinations in Large Language Models(LLM):Metrics and Mitigation

Thrilok. Kolla
Alliance School of Advanced Computing
Alliance University
Bengaluru, Karnataka
kollathrilok@gmail.com

Tina Babu
Alliance School of Advanced Computing
Alliance University
Bengaluru, Karnataka
tinababup@gmail.com

Rekha R Nair
Alliance School of Advanced Computing
Alliance Unniversity
Bengaluru, Karnataka
rekhasanju.sanju@gmail.com

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***Abstract*—Large Language Models' (LLMs') quick development has transformed natural language processing and made it possible to produce content that appears human in a variety of applications. One major issue that still exists, though, is the problem of hallucinations, in which LLMs generate information that is not supported by the incoming data or empirical knowledge. These hallucinations pose hazards in crucial applications, including the legal, medical, and educational fields, by compromising the quality and dependability of AI-generated content. Resolving this issue is essential for enhancing the reliability and practical applications of LLMs. Through a thorough approach that incorporates quantitative measures and visualization techniques, this study seeks to assess hallucinations in LLMs.The overall objective is to establish robust methods for identifying, quantifying, and comparing hallucination patterns across different models and applications. Through the scrutiny of the performance of different large language models (LLMs), the research seeks to determine patterns and causative factors of hallucinations. Developing more accurate and dependable models will be facilitated in the process. The findings of the investigation should enable us to better comprehend the mechanisms of hallucinations and develop effective mitigation strategies.**

***Index Terms*—BLEU Thresholds, retrieval-augmented generation (RAG), evaluation metrics, visualization methods, hallucinations, and natural language processing, Confusion Matrix, Semantic Similarity, and Factual Accuracy**

I. Introduction

The rapid development of large language models (LLMs) has revolutionized natural language processing across numerous applications and enabled one to generate content that sounds human. Hallucination, where LLMs report facts not backed by input data or empirical facts, is a serious problem that remains. These hallucinations can be dangerous in important areas such as the law, medicine, and education professions by diminishing the quality and reliability of AI text. This condition must be repaired to enhance the reliability and value of LLMs [1].

The methodology employed in this study integrates visualization techniques with quantitative measurements to assess hallucinations in LLMs. Developing scientifically sound techniques for identifying, measuring, and comparing hallucination tendencies across various models and domains is the main goal. In an attempt to pinpoint patterns and factors that lead to hallucinations, the study closely analyzes the performance of several LLMs. Models will be more accurate and dependable as a consequence. In addition to guiding the creation of workable mitigation strategies, the study should offer insightful information about the mechanisms underlying hallucinations [2].

In order to achieve these goals, the research applies a binary classification approach to identify hallucinations; the actual label tells whether the content is real or hallucinogenic. A confusion matrix to illustrate how the hallucination identification model separates the actual and hallucinated content is employed to test the performance of the model. The performance of the model at different levels of BLEU is tested with a range of measures, such as accuracy, recall, and F1-score, which serve as a hallucination filter parameter. These are steps taken to verify how efficiently the model is able to maintain false positives and true positives at an equal frequency [3].

The framework by which similarity metrics like BERTScore and ROUGE-L F1 are used to calculate the proximity between generated and reference text is examined in this study. Semantic similarity and coherence in the generated text can be more easily perceived through these metrics. Retrieval-augmented generation (RAG) is employed in the research to reduce hallucinations by anchoring the generated text with real pieces of a knowledge base. The integration of pertinent external data into the content generation process increases the factuality of artificial intelligence-generated content. Heatmaps, density plots, radar plots, scatter plots, bar and line plots, and other visualization tools are utilized in the paper to quantify hallucination ratings and semantic similarity between domains and models [4]

- In order to properly detect hallucinations in LLM output, develop a binary classification model.
- Examine the F1-score, recall, and precision of the hallucination detection at different BLEU thresholds.
- Two similarity measures that can be used to estimate how closely a generated and a reference text are ROUGE-L F1 and BERTScore.
- Factual accuracy can be improved and hallucinations reduced by using RAG to pull out factual passages from a knowledge source.
- There are some graphical aids to the comparison of semantic similarity and hallucination ratings across domains and models.

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II. Literature Survey

Recently, the impact of hallucinations in large language models (LLMs) on the accuracy and dependability of information produced by artificial intelligence has become prominent. Incorrect information and hallucinations happen when models produce output that is not based on input data or existing knowledge. Detection and mitigation systems for hallucinations have been proposed by most research works, highlighting the requirement for high-strength measurement and assessment methodologies. This literature review provides a clear overview of the state of the field currently through its breakdown of the findings and methodology of leading studies of the subject [5].

Binary classification models are the primary way of recognizing hallucinations as they label generated content based on pre-defined conditions as either real or hallucinatory. To measure the performance of these models and report data about classification correctness, confusion matrices are generally utilized. Algorithms for detecting hallucinations are commonly assessed using accuracy, recall, and F1-score metrics. The F1 score attempts to reconcile the proportion of true positives out of all real positives, expressed by a recall, and the proportion of true positives out of all positive predictions, expressed by accuracy. Typically employed as measures to distinguish hallucinations, these parameters are typically measured under different levels of BLEU [6].

Aside from binary classification, similarity scores are also essential in measuring the degree to which well-written text is similar to reference texts and still makes sense. Two such popular scores used in the literature are ROUGE-L F1 and BERTScore. ROUGE-L F1 measures overlap and coherence of the content by finding the longest common subsequence of the generated and reference text. By computing cosine similarity on contextualized embeddings, BERTScore gives a finer-grained understanding of semantic similarity. While researchers can measure the validity of AI-created content and the potential places where hallucinations might occur using these benchmarks [7].

Scholars have investigated how the use of Retrieval-Augmented Generation (RAG) enhances the accuracy of large language models (LLMs). RAG diminishes the risk of misinformation through incorporating factual pieces of information from a knowledge base in the material to be produced. It establishes a factual and contextual platform for content creation based on using outside data. RAG can significantly enhance the reliability of AI-generated content by reducing the number of wrong or psychoactive answers, as reported by various studies. Comparative studies are the most traditional way to measure the effectiveness of RAG; they compare models with RAG to models without RAG based on a variety of metrics [8].

It is highly important for LLMs to employ visualization methods to understand and minimize hallucinations. Semantic similarity and hallucination ratings between different models and domains are represented by researchers through density plots, scatter plots, bar and line charts, radar charts, and heatmaps. By highlighting areas that require improvement, such visualization provides insightful information regarding LLM performance and reliability. For instance, radar charts enable them to compare multiple metrics simultaneously, while heatmaps allow them to represent the concentration and distribution of hallucinations over models and domains. These techniques assist researchers in developing more accurate models by enabling them to identify trends and patterns in hallucination incidents [9].

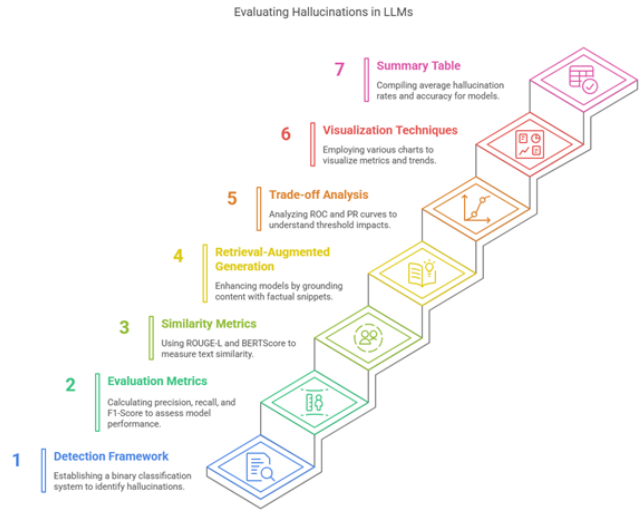

Fig. 1. Workflow of the Model

III. Methodology

Hallucinations in large language models (LLMs) are analyzed in this research through both quantitative and graphical approaches. The following subsections introduce techniques for detecting, quantifying, and comparing hallucination trends among models and types [10].

In Figure 1, the suggested model's methodology is shown.

A. Hallucination Detection Framework

Binary classification is used to identify hallucinations, in which:

- Correct label: $y \in \{0, 1\}$, where reality is represented by 0 and hallucination by 1.
- Anticipated Label: $\hat{y} \in \{0, 1\}$

A confusion matrix summarizes performance:

$$\begin{bmatrix} TP & FP \\ FN & TN \end{bmatrix} \tag{1}$$

Where:

- TP = true positives (correctly identified hallucinations)
- TN = true negatives (correctly identified factuals)
- FP = false positives (factual misclassified as hallucinated)
- FN = false negatives (hallucination misclassified as factual)

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B. Evaluation Metrics

1) Precision, Recall, and F1-Score:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}, \tag{2}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}, \tag{3}$$

$$\text{F1-Score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{4}$$

These metrics are evaluated under varying BLEU thresholds, which are treated as a filtering criterion for identifying hallucinations [11].

C. Similarity Metrics

1) ROUGE-L F1: Measures longest common subsequence between reference and generated text:

$$\text{ROUGE-L}_{F_1} = \frac{(1+\beta^2) \cdot \text{ROUGE-L}_{\text{precision}} \cdot \text{ROUGE-L}_{\text{recall}}}{\text{ROUGE-L}_{\text{precision}} + \beta^2 \cdot \text{ROUGE-L}_{\text{recall}}} \tag{5}$$

Where $\beta = 1$ for equal weighting.

2) BERTScore: Computes cosine similarity over contextualized embeddings:

$$\text{BERTScore}_{F_1} = \frac{1}{N} \sum_{i=1}^{N} \max_j \cos\left(e_i^{\text{gen}}, e_j^{\text{ref}}\right) \tag{6}$$

Where:

- e_i^{gen}: embedding of generated token i
- e_j^{ref}): embedding of reference token j

D. Retrieval-Augmented Generation (RAG)

To reduce hallucination, the model is enhanced using RAG, which retrieves factual snippets $R = \{r_1, r_2, \ldots, r_k\}$ from a knowledge base to ground the generated content G.The hallucination count is computed as:

$$H = \sum_{i=1}^{n} 1\{G_i \notin R\} \tag{7}$$

Where 1 is the indicator function.

E. Trade-off Analysis: ROC and PR Curves

To analyze BLEU-based thresholding, we model threshold $\theta \in [0, 1]$ and evaluate:

- ROC Curve: plots TPR (Recall) vs. FPR

$$FPR = \frac{FP}{FP + TN} \tag{8}$$

- PR Curve: plots Precision vs. Recall as θ varies.

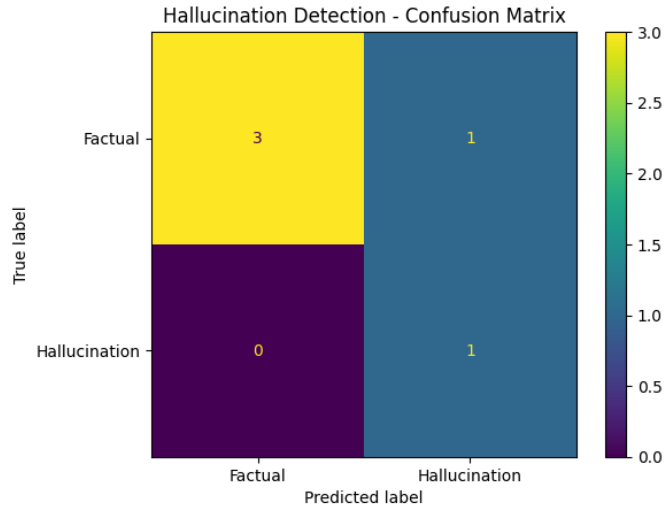

Fig. 2. Confusion Matrix

F. Comparative Visualization Techniques

Metrics are visualized using:

- **Bar and Line Charts**: To compare hallucination scores and semantic similarity.
- **Scatter Plots**: To relate factual accuracy to hallucination rate.
- **Radar Charts**: For multi-metric model comparison.
- **Heatmaps**: To show distribution and concentration across models/domains.
- **Density Plots**: to model probabilistic hallucination distribution $f_H(h)$.

IV. Results and Discussion

This confusion matrix visualizes the performance of a hallucination detection model. The matrix compares predicted labels (Factual or Hallucination) with true labels. The model correctly predicted 3 factual cases (top left) and 1 hallucination case (bottom right). It misclassified 1 factual case as a hallucination (top-right) and made no errors in labeling hallucinations as factual (bottom-left). Overall, the model shows strong performance in identifying hallucinations but has a slight tendency to label factual content as hallucinated incorrectly. The color gradient indicates the frequency of predictions, with brighter colors representing higher counts [12].

This bar chart displays ROUGE-L F1 scores for five text samples, measuring how well each generated text matches a reference by capturing the longest common subsequence. Higher scores indicate better content overlap and fluency. Sample 2 and Sample 5 achieved the highest score of 1.0, suggesting perfect or near-perfect matches. Sample 1 also performed well with a score above 0.85. Sample 3 and Sample 4 had lower scores, around 0.67 and 0.57 respectively, indicating weaker similarity or coherence. The chart highlights variability in performance across samples and may suggest inconsistency in the generation quality or relevance of some outputs [13].

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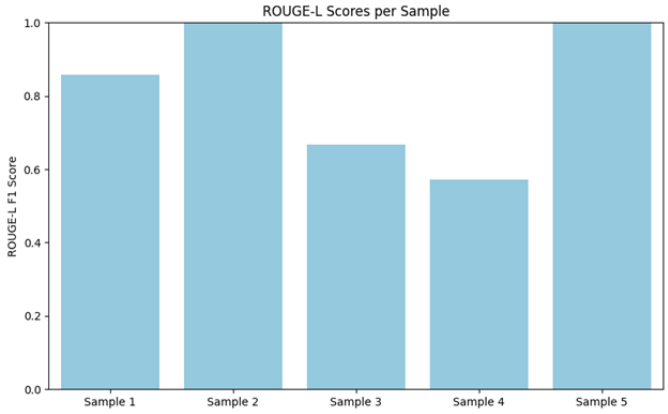

Fig. 3. Bar Chart

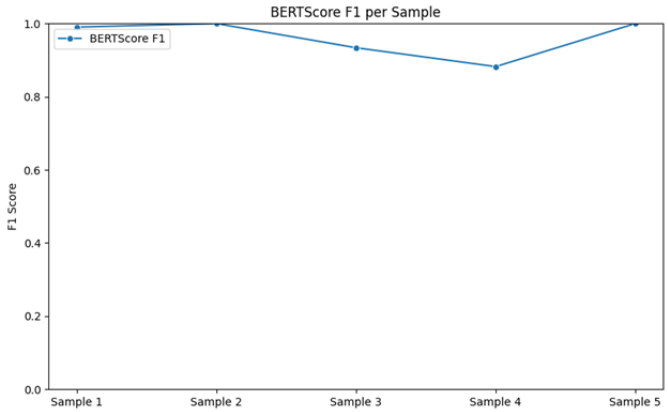

Fig. 4. BERT score

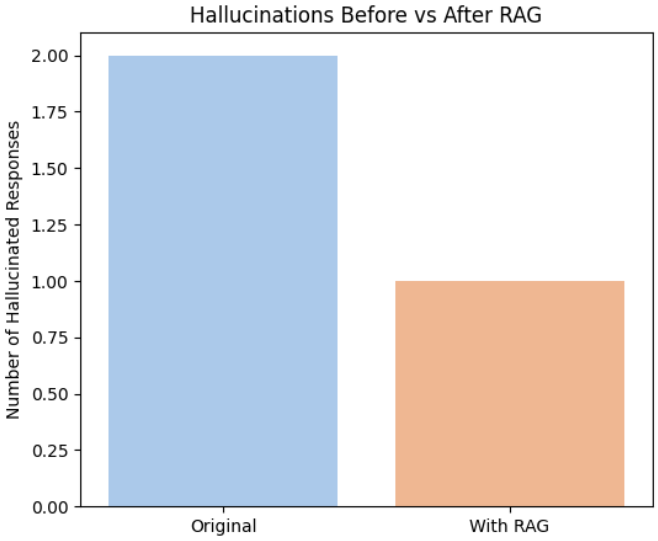

Fig. 5. RAG

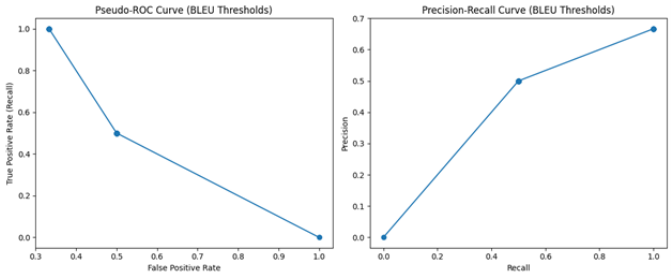

Fig. 6. Bleu Thresholds

This line graph shows BERT Score F1 values for five text samples, assessing semantic similarity between generated and reference texts using contextual embeddings. Scores are close to 1.0 across all samples, indicating strong semantic alignment. Samples 1, 2, and 5 achieved perfect or near-perfect scores (1.0), suggesting excellent semantic preservation. Sample 3 and Sample 4 have slightly lower scores around 0.93 and 0.89 respectively, indicating minor deviations in meaning. The overall high scores reflect good quality in generated texts, with minimal semantic drift, and the BERT Score's sensitivity to meaning beyond surface-level text is effectively demonstrated through these consistent evaluations [14].

This bar chart compares the number of hallucinated responses generated by a language model before and after applying Retrieval-Augmented Generation (RAG). The "Original" model produced 2 hallucinated outputs, while the model "With RAG" reduced hallucinations to just 1. This indicates a 50% reduction in factual inaccuracies when RAG was incorporated. RAG enhances output by retrieving relevant external information during generation, which likely contributes to better grounding and factual consistency. The chart visually supports the conclusion that integrating retrieval mechanisms can significantly improve the reliability of AI-generated content by minimizing hallucinated or fabricated responses [15].

These plots show the effect of BLEU score thresholds on classification performance. The left plot is a pseudo-ROC curve, illustrating the trade-off between true positive rate (recall) and false positive rate as the threshold changes. It shows that increasing the threshold reduces recall but also reduces false positives. The right plot is a precision-recall curve, highlighting the balance between these two metrics. As recall increases, precision also improves, peaking at a recall of 1.0. Together, these curves help assess how BLEU thresholds influence the system's ability to distinguish between correct and incorrect outputs, guiding optimal threshold selection for evaluation [16].

Using various BLEU levels for classification, this plot shows how precision, recall, and F1 scores change. Lower thresholds yield a consistent F1-score of about 0.5 due to modest precision and recall. Recall jumps to 1.0 and precision climbs to roughly 0.67 as the threshold is raised to 0.8, bringing the F1-score to 0.8. An area of constant F1 score is represented by the threshold of 0.4 (dashed line), which is chosen to balance performance across metrics. For evaluation or filtering in a language production task, this threshold probably provides an acceptable trade-off between detecting correct outputs (recall)

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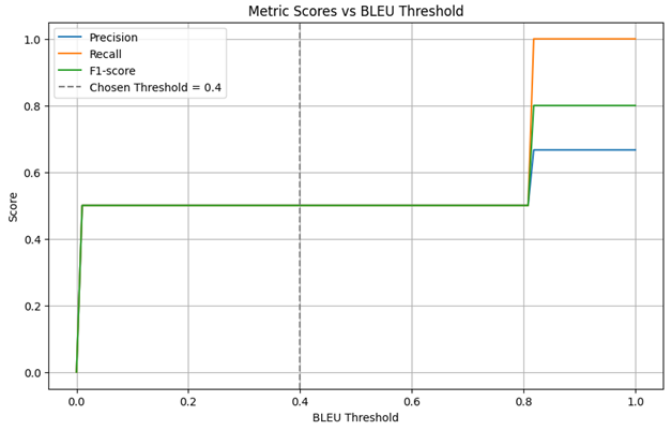

Fig. 7. Metric Score vs Bleu Thresholds

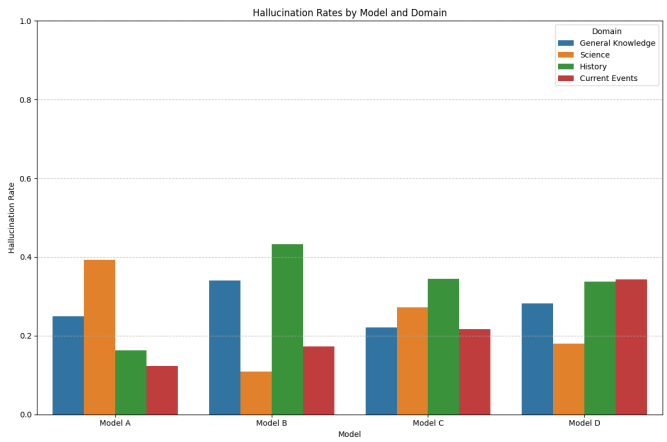

Fig. 8. Hallucination Rates by Model and Domain

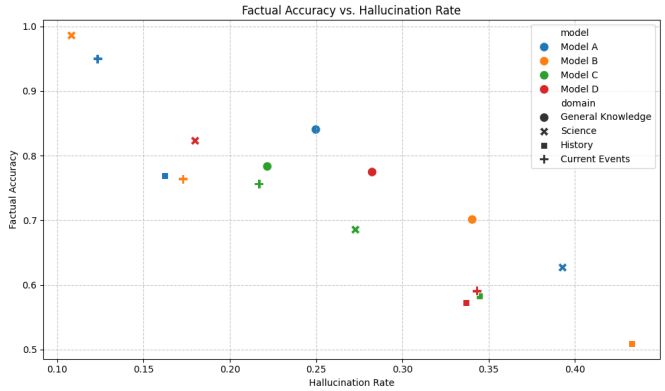

Fig. 9. Factual Accuracy vs Hallucination Rate

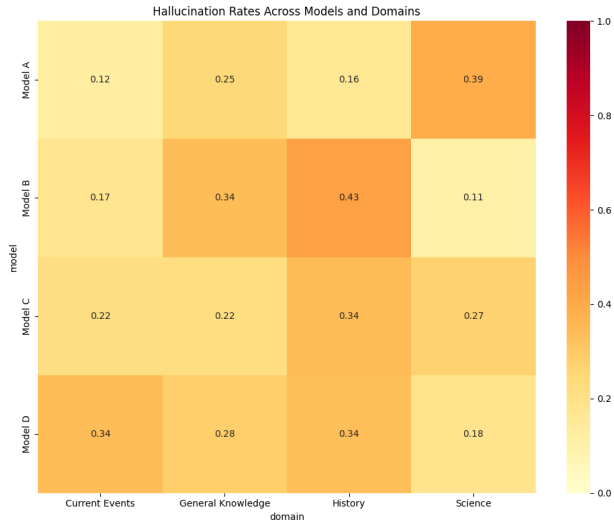

Fig. 10. Hallucination Rates Across Models and Domains

and reducing false positives (accuracy) [17].

It is a bar chart with the caption ”Hallucination Rates by Model and Domain.” Four distinct models (Model A, Model B, Model C, and Model D) are compared for their hallucination rates in four different domains: current events (red), science (orange), history (green), and general knowledge (blue). For history, Model A has the highest hallucination rate, whereas for general knowledge, it has the lowest. Current events have a high hallucination rate, but general knowledge has a low rate, according to Model B. All domain rates are comparatively balanced in Model C, with Science showing somewhat higher rates. Model D displays balanced rates as well, however the rates in History are little greater. [18].

It is a scatter plot with the title ”Factual Accuracy vs. Hallucination Rate.” The Hallucination Rate (x-axis) ranges from 0.1 to 0.35, and the Factual Accuracy (y-axis) ranges from 0.5 to 1.0. The following symbols and colors stand for different models and domains: Model A (blue cross), Model B (orange circle), Model C (green plus sign), and Model D (red x). The following domains are represented by black symbols: Current Events (black x), History (black plus sign), Science (black circle), and General Knowledge (black cross). To show the relationship between factual correctness and the hallucination rate, each point on the plot represents how well a certain model performs in a given domain. Lower factual accuracy is typically linked to increased hallucination frequencies. [19].

The heatmap depicted here is called ”Hallucination Rates Across Models and Domains.” It displays the frequencies of hallucinations for four models (Model A, Model B, Model C, and Model D) in four different domains: science, history, general knowledge, and current events. The intensity of the hue indicates the frequency of hallucinations; higher rates are indicated by darker colors. Model B has a lower hallucination rate of 0.11 in the same domain as History, whereas Model A has a higher rate of 0.39. Model D consistently scores 0.34 in Current Events, while Model C displays a high rate of 0.35 in General Knowledge. [20].

Four density charts that depict the distribution of hallucination rates for the four distinct models—Model A, Model B, Model C, and Model D—are displayed in the image. ”Density” is plotted on the y-axis and ”Hallucination Rate” is plotted on

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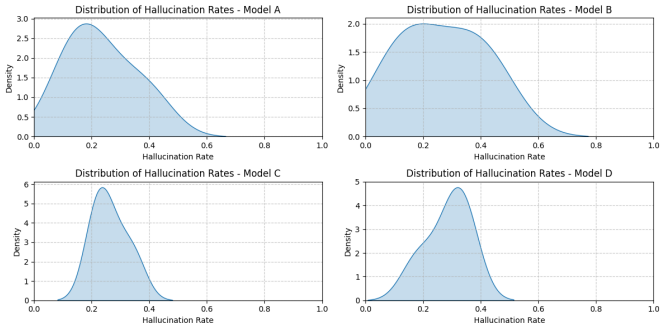

Fig. 11. Hallucination Distributions

the x-axis. [21].

- The plot of Model A peaks at a hallucination rate of 0.2 and then progressively declines.
- Although Model B's plot resembles Model A's in shape, it has a greater peak at about 0.3.
- The distribution in Model C's figure is more constrained, peaking at about 0.1.
- The plot of Model D displays a broader distribution, with a peak that extends farther along the x-axis and is likewise approximately 0.1.

The frequency of various hallucination rates within each model's predictions or outputs is depicted by these density charts. By showing that Model B typically has greater hallucination rates while Model C and D typically have lower rates, they aid in visualizing the dependability of each model. Understanding and enhancing AI model performance requires knowledge of this information. [22].

It's a radar graphic with the headline "Radar Chart: Normalized Hallucination vs. Factual Accuracy." In terms of factual correctness and hallucination rate, it contrasts four models (Model A, Model B, Model C, and Model D). Models A, B, C, and D are, respectively, blue, orange, green, and red. Every model is represented by a distinct color.The concentric circles show values in increments of 0.2 from 0.0 to 1.0. Each model's performance is shown in the radar image in relation to these metrics. Consider, for example, a model with low hallucination rate and great factual accuracy [23].

V. Conclusion

By establishing a comprehensive framework for measurement, detection, and cross-domain comparison, the serious issue of hallucinations in LLMs is successfully addressed. To identify and treat hallucinations, the study uses similarity metrics, binary classification, and retrieval-augmented generation (RAG). The model's performance is reliably evaluated using the evaluation metrics, F1-score, precision, and recall. Additional useful information about the correctness and dependability of AI-generated data can be found in visualization tools.

According to the study's findings, in order to increase the credibility of Large Language Models (LLMs), created content

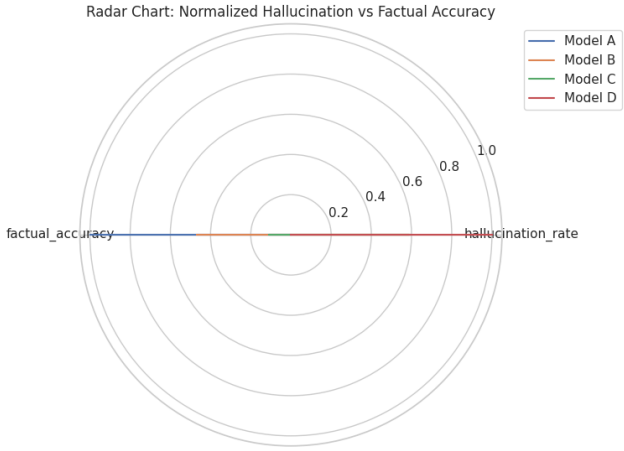

Fig. 12. Normalized Hallucination vs Factual Accuracy

must be firmly rooted in data. By carefully analyzing hallucination patterns and methods for reducing them, the research helps to build more reliable and accurate models. Future developments in natural language processing are expected to be guided by the knowledge gained from this study, allowing LLMs to generate outputs that are true, accurate, and internally consistent. To limit their use in real-world situations and successfully handle the issues brought on by hallucinations, it is imperative to continuously evaluate and improve LLMs in light of these discoveries.

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